MIL-M-36755B
2 October 1975
SUPERSEDING
MIL-M-0036755A(DSA-DM)
22 May 1972 and
MIL-M-36755
3 May 1971

MILITARY SPECIFICATION

METER, HYDROGEN ION TEST

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high accuracy portable battery and line-operated pH meter, suitable for use in precise laboratory investigations, studies involving measurement of oxidation-reduction potentials, titrations and special measurements.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-636	Boxes, Shipping, Fiberboard.

Military

MIL-P-116	Preservation-packaging, Methods of.
MIL-M-7298	Manual, Technical, Commercial Equipment.

FSC 6630

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73.11	Dimensions of Plugs and Receptacles - General Purpose, 125 Volts, 15 Amperes, 2-Pole, 3-Wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout, and shall be as specified herein. Material not specified shall be of good commercial quality for this type of equipment.

3.2 Design and construction. The instrument shall consist of a meter, battery, an electrode holder assembly, carrying case (if applicable), a handle, a three-conductor line cord and accessories. The meter (less handle) shall have nominal dimensions of 12 inches wide by 8 inches high by 5 inches deep and shall weigh not more than eight pounds (less battery and accessories). The meter shall have a battery recharging system and shall have provision to automatically maintain a battery charge when in operation on AC. The meter shall be stable in use and shall be capable of operating in vertical, horizontal or tilting positions.

3.2.1 Meter. The meter shall be all solid state, manually temperature-compensated, and of either a null-balance or direct reading type. The meter shall have a stabilized power supply and amplifier circuitry. The meter shall be fitted with readily accessible, convenient-to-use controls and shall include a power selector switch to permit battery operation, AC line operation or recharge of the battery.

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3.2.1.1 Scales. The meter shall have a dual reading scale. One shall indicate measurement in full range of 0 to 14 pH units with smallest scale division not more than 0.05 pH. The other scale shall indicate from 0 to at least plus and minus 700 millivolts (mv.), with the smallest scale division not more than 10 mv. Readout dial shall be observable through a cover of clear methyl-methacrylate resin.

3.2.1.2 Temperature compensation. Temperature compensation shall cover the range of 0^{0} to 100^{0} centigrade (^{0}C.) and shall be operated by means of a knob. A scale shall be provided and marked with 2^{0}C. divisions.

3.2.2 Battery. The battery shall be a rechargeable, nickel cadmium type, rated at 14.4 volts and 500 milliampere hours. It shall be capable of providing not less than 35 hours of operation per charge.

3.2.3 Electrode holder assembly. The electrode holder assembly shall consist of a rod and bracket for holding two electrodes. The rod shall be corrosion-resisting steel, 9-3/16 ± 13/16 inches long and 15/64 to 25/64 inch in diameter, and shall be secured to a corrosion-resisting metal post on the right side of the meter. The bracket shall be made of corrosion-resisting steel, shall be readily securable to the rod and shall provide side-to-side movement with vertical adjustment capability. As an alternate, the bracket shall be made of polypropylene resin.

3.2.4 Storage of accessories.

3.2.4.1 Carrying case. The carrying case shall be black leather. It shall have provisions to contain the meter, components and accessories with cushioning for the delicate accessory items. It shall also have a suitable locking device.

3.2.4.2 Storage compartment. As an alternate to the carrying case (see 3.2.4.1), the meter housing shall have a storage compartment to contain the components and accessories. It shall have cushioning for the delicate accessory items and a suitable locking device to effect closure of the compartment.

3.2.5 Handle. The handle shall be swing type or integral with the meter case. The integral handle shall be constructed of oxidized aluminum and shall be 3/4 ± 1/8 inch wide and 7/8 ± 1/8 inch thick. The swing type handle, 1 ± 1/8 inch wide, 0.091 inch (11 B & S gauge) thick, shall be fabricated of chromium-plated carbon steel and shall be designed to facilitate transportation or operation of the instrument in a vertical, horizontal or tilted position. The handle shall be capable of being readily secured so as to support the meter in a tilted position (45 ± 5^{0} to the horizontal). The meter case with integral handle shall be so designed that the meter shall be capable of being tilted to 45 ± 5^{0} to the horizontal.

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3.2.6 Accessories. Accessories shall consist of the following:

a. One calomel electrode with 30 inch lead.
b. One glass tri-purpose electrode with 30 inch lead.
c. One combination electrode with 30 inch lead.
d. One bottle (not less than 25 ml.) of saturated potassium chloride solution.
e. One bottle (not less than 25 ml.) of buffer solution or tube of buffer powder, 4.01 pH.
f. One bottle (not less than 25 ml.) of buffer solution or tube of buffer powder, 7.00 pH.
g. One bottle (not less than 25 ml.) of buffer solution or tube of buffer powder, 9.0 - 9.2 pH.

3.2.7 Line cord. Line cord shall be 7 ± 1 feet in length, flexible, detachable, type SJ or SJT, No. 18 AWG, three-conductor cord. The input end of the cord shall terminate in a parallel-blade attachment plug with grounding pin. The meter end of the cord shall terminate in a parallel-slot plug with grounding insert. Both connectors shall be in accordance with the mating features of ANSI Standard C73.11.

3.2.8 Feet. There shall be four feet on the bottom plate of the meter positioned near each corner to effect stability on a plane flat surface.

3.3 Electrical characteristics. The meter shall be capable of operating on 105-125 volt, 50-60 Hertz (Hz), AC power. By effecting a simple modification, the instrument shall be capable of operating on 210-250 volt, 50-60 Hz, AC. Procedure to convert to 210-250 volts AC shall be included in operating manual (see 3.7).

3.3.1 Line cord continuity. Contacts on the connector at the input end of the line cord shall be connected to the corresponding contacts on the connector at the meter end.

3.3.2 Grounding circuit resistance. Resistance from the grounding pin on the line cord input connector to the exposed metal parts of the meter shall not exceed 150 milliohms when measured as specified in 4.4.2.

3.3.3 Dielectric strength. The instrument shall withstand the dielectric test specified in 4.4.3.

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3.4 Performance.

3.4.1 Meter.

3.4.1.1 Millivolt scale accuracy. The meter shall comply with the millivolt scale accuracy test specified in 4.4.4.1.1.

3.4.1.2 pH scale accuracy. The meter shall comply with the pH scale accuracy test specified in 4.4.4.1.2.

3.4.1.3 pH accuracy. The meter shall be accurate to ± 0.04 pH over a span of 3 pH from any standardization point when tested as specified in 4.4.4.1.3.

3.4.2 Electrodes. All electrodes furnished with the meter shall operate without failure when tested as specified in 4.4.4.2.

3.5 Finishing and coating. All surfaces shall have a smooth finish and shall be coated in accordance with the manufacturer's standard practice. Unless otherwise specified, all metal parts shall be suitably coated or treated to resist corrosion.

3.6 Identification marking.

3.6.1 Meter. Each meter shall have a nameplate of corrosion-resisting metal, securely mounted in a conspicuous location, and permanently and legibly marked with the following information:

Manufacturer's name or registered trademark
Model number
Serial number
Volts
Frequency
Watts

3.6.2 Control information. All control identification and position settings shall be permanently and legibly marked.

3.6.3 Carrying case (if applicable). The leather case shall be permanently marked with the manufacturer's name or registered trademark.

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3.7 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.8 Workmanship. Workmanship shall be first class throughout. The meter, including components and accessories, shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I. The unit of product for examination purposes shall be the complete instrument as specified in 3.2.

Table I. Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Table II. The Unit of product for test purposes shall be the complete instrument as specified in 3.2. Acceptance number shall be zero.

Table II. Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level
Line cord continuity	3.3.1	4.4.1	100%
Grounding circuit resistance	3.3.2	4.4.2	100%
Dielectric strength	3.3.3	4.4.3	100%
Millivolt scale accuracy	3.4.1.1	4.4.4.1.1	100%
pH scale accuracy	3.4.1.2	4.4.4.1.2	100%
pH accuracy	3.4.1.3	4.4.4.1.3	100%
Electrodes	3.4.2	4.4.4.2	100%

4.3 Examination.

4.3.1 Instrument. The instrument shall be examined to determine compliance with requirements contained in this specification. The instrument shall be examined for possible defects, including, but not restricted to, those listed in Table III.

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Table III. Classification of defects

Classification	Defect
Major	
101	Components missing.
102	Identification marking on meter or carrying case (if applicable) missing, incorrect or incomplete.
103	Meter or electrode holder assembly not free of sharp edges.
104	Meter not stable on plane flat surface.
105	Readout or dial windows not clear.
106	Readout or dial windows not free of scratches, cracks and warpage.
107	Dial face not free of warpage, fading, and discoloring.
108	Null indicating scale not clear or not centrally located on dial face (if applicable).
109	Coating not free of chipping and peeling.
110	Surfaces not completely coated.
111	Metal parts not free of corrosion.
112	Electrodes not free of scratches, etched marks and cracks.
113	Electrodes not securely held by electrode bracket holder.
114	Locking device of carrying case or storage compartment not functioning.
115	Leather case not free of tears, rips and gouges (if applicable).
116	Meter in body of case or items on inside of cover not held securely in place (if applicable).
Minor	
201	Leather case not free of scratches, scuffing and mars (if applicable).
202	Coating on meter not free of wrinkles, smudges, scratches and mars.
203	Feet not secure (if applicable).

4.3.2 Dimensional examination. The meter components and accessories shall be examined for defects in dimensions. Any dimensions not within the tolerance specified herein shall be classified as a defect.

4.3.3 Preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

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Table IV. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence or method of application.

4.4 Tests.

4.4.1 Line cord continuity. Continuity shall be checked from the contacts on the connector at the input end of the line cord to the corresponding contacts on the connector at the meter end (see 3.3.1).

4.4.2 Grounding circuit resistance. With the line cord connected to the instrument, resistance shall be measured from the grounding pin on the line cord input connector to all exposed metal parts of the meter by means of a resistance measuring device with ± 2% accuracy.

4.4.3 Dielectric strength. An AC potential of 900 volts shall be applied between the current carrying parts and the metal enclosure. The test shall be conducted with a suitable transformer with means for regulating the output voltage. Starting from zero, applied potential shall be increased gradually until 900 volts is reached and maintained for one minute (see 3.3.3).

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4.4.4 Performance.

4.4.4.1 Meter. In meter performance tests listed below, a known variable potential shall be used, as supplied from a calibrated potentiometer (millivolt source) and shall be applied through a resistance of 500 megohms to the input terminal of the pH meter. The positive lead from the potentiometer shall be connected to the glass electrode jack on the pH meter. The 500 megohm resistor shall be shielded to avoid capacity pickup. Meter shall be allowed to warm up thoroughly before testing to insure stability.

4.4.4.1.1 Millivolt accuracy.

a. Compare the millivolt scale with the calibrated potentiometer at a voltage of 100.0 mv.
b. When the millivolt scale of the meter reads exactly 100.0 mv. the potentiometer shall read 100.0 ± 1.0 mv.
c. Increase the meter readings in 100.0 mv. steps with the calibrated potentiometer. The reading on the potentiometer shall increase by 100.0 ± 5.0 mv. for each step for the entire range.

4.4.4.1.2 pH scale accuracy (millivolt sources). Determine the difference in voltage on the pH scale of the meter from pH 1 to pH 2, pH 6 to 7, pH 9 to 10, using the calibrated potentiometer. The difference in voltage of one pH unit shall be 59.16 ± 1.00 mv.

4.4.4.1.3 pH accuracy. Standardize the pH meter at pH 7 (at 25° C.) as per manufacturer's instructions, using the calibrated potentiometer (millivolt source) connected to the input of pH meter. Compare difference in voltage at pH 4 from the standardization point (pH 7); repeat at pH 10. The difference in voltage of 3 pH units shall be 177.468 mv. ± 2.370 mv.

4.4.4.2 Electrodes. Each electrode shall be tested to assure that it is in operable condition and is suitable for use with pH meters in determining the pH of solutions.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Level A. Each meter along with accessories, excluding the battery, shall be positioned within their respective compartments in the carrying case. The case with contents shall be secured. The battery shall be packaged in a paperboard box of appropriate size and the box inserted into a heat-sealed polyethylene bag. The case with components and the packaged battery shall be packaged in accordance with MIL-P-116, method IC-2.

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5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A. Each unit, packaged as specified in 5.1.1, shall be packed in an exterior container designed for a Type 1 load and constructed in accordance with PPP-B-636, Class weather-resistant, Grade V3c or V3s, Style RSC. Closure and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Each unit, packaged as specified in 5.1.1, shall be packed in an exterior container constructed in accordance with PPP-B-636, Type CF, Class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following item shown in the Federal Supply Catalog:

National Stock No.	Item Identification
6630-00-105-8648	METER, HYDROGEN ION TEST, Field, Battery or Line Powered, 115 Volt, 50-60 Hz, AC

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Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6630-0240
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-M-36755B METER, HYDROGEN ION TEST

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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